Lab 1

An assignment for class **MET CS579** Database Management at Boston University due on **February 9, 2023**, instructed by **Jae Lee**. Students Jin Yu Zhang and Wei Wang submitted on **February 8, 2023**.

Assumptions

Description for things not included in the requirements:

- Customer is not an entity since it doesn't have any attribute, and we assume that each customer has only one headquarter by the definition of a headquarter. Therefore, customer and `HEADQUARTER` are equivalent here
- Each customer has one or more branch stores. We assume that there is no limit for the number of `Branch_Store` for a `HEADQUARTER`
- According to the requirements, we should keep the information about the retail store who issued the order for an order. However, since there is already a relationship between `ORDER` and `HEADQUARTER`, we will not put this as an attribute in `ORDER`
- According to the requirements, we should keep the information about the employee(s) of the Boston Shoes who handle the order for an order. However, since there is already a relationship between `ORDER` and `EMPLOYEE`, we will not put this as an attribute in `ORDER`. Also the requirement says an employee handles at most four order, we assume an employee is allowed to handle no orders
- There are not limits for how much suborders a `BRANCH_STORE` can be shipped to, and how many suborders can a `SHOES_TYPE` hold. There are hence form with `N` (many) constraint in this assignment
- We assume an `EMPLOYEE` come with only one classification

Questions

Known issues:

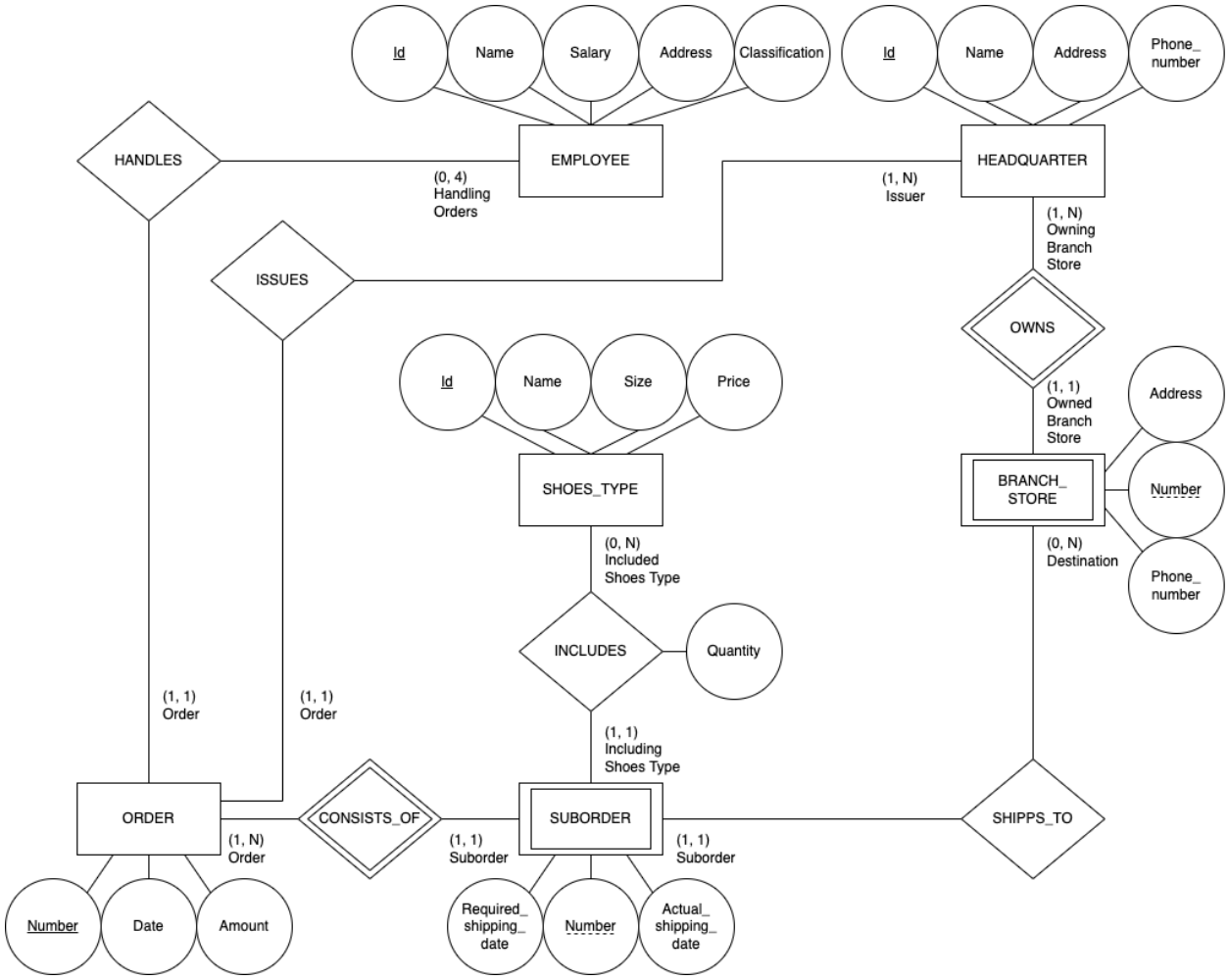
- Is sequence number a derived value? If it is, how do we define the partial key for `SUBORDER` and `BRANCH_STORE`?
- 5-2, 5-4, and 6-2. Should we store them as attributes or relationship types are enough?

Entity Types

Name	Description	Attribute			
		Name	Description	Multi-Values	Derived
EMPLOYEE	Employees works for Boston Shoes.	Id {K} Name Salary Address Classification		N N N N N	N N N N N
SHOES_TYPE		Id {K} Name Size Price		N N N N	N N N N
ORDER		Number {K} Date Amount		N N N	N N N
SUBORDER		Number {WK} Required_shipping_date Actual_shipping_date	Number is a sequence number starting with 1 and is only unique among a certain ORDER.	N N N	N N N
HEADQUARTER	The customers of Boston Shoes such as Walmart, Target, and Kohls.	Id {K} Name Address Phone_number		N N N N	N N N N
BRANCH_STORE		Number {WK} Address Phone_number	Number is a sequence number starting with 1 and is only unique among a certain BRANCH_STORE.	N N N	N N N

Relationship Types

Name	Description	Participating Entity Types		Attribute
		Name	Structural Constraint	
OWNS	Each customer (headquarter) has one or more branch stores.	HEADQUARTER BRANCH_STORE	(1, N) (1, 1)	None
ISSUES	An order is issued by a retail store headquarter.	HEADQUARTER ORDER	(1, N) (1, 1)	None
HANDLES	Boston Shoes employees handle the order.	EMPLOYEE ORDER	(0, 4) (1, 1)	None
CONSISTS_OF	An order consists of one or more suborders.	SUBORDER ORDER	(1, 1) (1, N)	None
SHIPS_TO	A suborder is shipped to exactly one branch store.	SUBORDER BRANCH_STORE	(1, 1) (0, N)	None
INCLUDES	A suborder includes one or more shoe types.	SUBORDER SHOES_TYPE	(1, N) (0, N)	Quantity


Id
Name
Salary
Address
Classification
Id
Name
Address
Phone_ number
HANDLES
EMPLOYEE
(0, 4) Handling Orders
HEADQUARTER
(1, N) Issuer
(1, N) Owning Branch Store
ISSUES
OWNS
Id
Name
Size
Price
(1, 1) Owned Branch Store
Address
SHOES_TYPE
BRANCH_ STORE
Number
(0, N) Included Shoes Type
(0, N) Destination
Phone_ number
INCLUDES
Quantity
(1, 1) Order
(1, 1) Order
(1, 1) Including Shoes Type
ORDER
CONSISTS_OF
SUBORDER
SHIPPS_TO
(1, N) Order
(1, 1) Suborder
(1, 1) Suborder
Number
Date
Amount
Required_ shipping_ date
Number
Actual_ shipping_ date